



Department
of Statistics

ST498 Capstone Project
Guidelines for MSc Data Science
Students

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The Capstone Project Overview

The Capstone is a one-unit core course undertaken by all MSc in Data Science students. It requires groups of students to carry out research and analysis to address a real-world data science problem that is relevant to a partner organisation. The Capstone allows students to extend their knowledge and skills and apply disciplinary and methodological knowledge and skills they learned in the MSc in Data Science courses to address a real-world data science problem. The Capstone partners include public sector bodies, private sector companies and international organisations.

As its learning outcome, the Capstone envisages that students will learn how to work on a data science project, including a variety of activities such as:

- understanding research questions posed by the partner
- formulating their research questions
- identifying and gathering needed data
- cleaning and processing data
- conducting exploratory data analysis
- applying machine learning and statistical methodologies that are best suited to address the underlying research questions
- learning new methodologies as required by the project
- applying the most suitable software technologies for data processing
- using modern data science tools
- practising how to efficiently communicate research results to project partners, end users, data science experts, and a general audience.

The Capstone is a collaborative project. Students work in groups and each project is supervised by a Department of Statistics faculty, who provides advice and monitors progress. (See [The Role of LSE Supervisors](#)). Each group is required to submit a Capstone report and an individual report by the date indicated (see the Section [Capstone Timeline](#)).

Allocation of Capstone Projects to Students

Capstone Projects are normally assigned to students by Week 5-6 of the Autumn Term. The students are required to form groups of a given size and assert their individual preferences for the offered project topics. The allocation of projects to groups of students is based on using a random maximum weight matching procedure that uses input student preferences (aggregated per project).

Capstone Timeline

Project progress updates and feedback are conveyed through i) individual team meetings with the LSE supervisor and the company partner, ii) all-hands meetings presenting the project progress report to the entire class, and iii) workshops in which Capstone project results are featured (e.g. in a poster session).

Date	Event
Winter Term Week 2 (approx.)	First all-hands meeting (including a 10-minute presentation of project goals and work plan)
Winter Term Week 10 (approx.)	Second all-hands meeting (project update)
Spring Term Week 4 (approx.)	Submit first draft of report
6 August 2027	Submission of final project report, including individual report document.
Approx one week after submission	Students contacted with request to complete Capstone evaluation form

It is important to note that you are expected to work continuously on your Capstone Project throughout the year. This includes attendance at all of the above meetings. If required by your Capstone partner, you are also expected to make yourself available for any meetings during the Spring vacation period. This is per the School's [Conditions of Registration](#), specifically point 3 therein.

The Capstone Group Report

Submission

Capstone Projects are submitted via the submission portal on the [ST498 Moodle Page](#). The requirements for submission are as follows:

1. **One** group member needs to upload the report to Moodle on behalf of the group by the deadline. In addition, please have an electronic copy of your report in a GitHub repository if you use one for your project. Please also add a working copy of all code and data (except datasets that are protected or have any other restrictions).

All Capstone Project reports and code will be uploaded onto Turnitin, the School's plagiarism detection software. Further details will be provided in an email from the MSc Programmes Administrator in advance of the submission date.

At the time of submission, you will be asked to grant your permission for your Capstone project report to be used as an exemplar for future MSc Data Science students.

Please note:

- You must include in an appendix within your report a copy of the final Terms of Reference (ToR) that was agreed with your partner (when applicable).
- Your submitted work must not contain the names of your groups – only candidate numbers should be used. You can produce a separate copy of the report containing your names for your partner if you wish.

The School regulations for late submission are as follows:

1. If a student believes that s/he has good cause not to meet the deadline (for example illness), s/he should first discuss the matter with their Capstone supervisor and/or the Capstone Project course coordinator (currently Dr Marcos Barreto) and submit a formal extension request. Normally extensions should only be granted where there is a good reason backed by supporting evidence (for example a medical certificate). Please see further information and the Extension Request form [here](#).
2. Any extension should be confirmed in writing to the student.
3. If a student fails to submit by the set deadline (or extended deadline as appropriate), five percentage marks will be deducted per day (this includes weekends, bank holidays and School closure days), or part, thereafter late.

See the [Academic Discipline and Academic Misconduct](#) page on the School website for more information.

Presentation and Layout of your Group Report

The elements described below are required parts of the report.

- **Title page:** this should include "Department of Statistics 2024/2025", the title of the Capstone Project, your candidate number, and the statement "Submitted for the Master of Science, London School of Economics, University of London". **Marking is done anonymously; please do not include your name on any part of the report.**
- **Acknowledgements** (optional)
- **Table of contents:** This is made up of three elements each giving page numbers:
 - Table of contents: an ordered list of the sections and subsections of the project (include the bibliography and any appendices);
 - List of figures: an ordered list of diagrams;
 - List of tables: an ordered list of tables.
- **Executive summary:** a summary of the work in no more than 1500 words (not more than 3 pages). This is an integral component of the Capstone report. The executive summary should provide a succinct and at the same time thorough overview of the main findings and recommendations. It should enable the reader to become acquainted with the content of the capstone report without having to read it all. The executive summary must conform to

the same formatting requirements as the rest of the capstone report. The executive summary should articulate what is your project about, why is the research problem a problem (in other words, why is it non-trivial to solve your problem), what are your main findings, and what are the practical implications of your results (e.g. guidelines for any end users of your work).

- **Introduction:** an introductory section that should include the aims of the study and a brief outline of the main sections of the capstone project. The introduction contains details about the research topic and the research questions, a summary of the main results, and a brief overview of the structure of the report. The introduction should not exceed 20 per cent of the entire report. It can be divided into sub-chapters:
 - Objectives and research questions: this section sets the context, outlines the main objectives and introduces the research questions that are addressed in the report.
 - Summary of main results: this part provides a succinct summary of the main results of the report.
 - Outline: a brief overview of the chapters of the report.
- **Main sections:** This part needs to introduce the research problem studied in the report more formally and in more depth. This section also needs to include a discussion of the methodology used to address the research questions studied and provide justification and rationale for choosing any specific methodologies. This part also needs to introduce the data sources used in the project and provide basic information about these data sources. This section needs to include a review of the literature, a description of the methods used and the results of the study.
- **Conclusion:** This section should contain conclusions that can be drawn from your work, possible extensions and further work. This section must synthesise the major findings from the previous chapters. The objective of this task is not to merely repeat the main findings, but to tie together the findings, analysis and interpretation and generate a conclusive statement. The conclusion must highlight only the significant outcomes of the analysis. It serves as a foundation for the guidelines which will follow.

The key outcome of the capstone report is to generate guidelines for the end users. The end users may be a company partner who proposed the research problem in the first instance, or any other potential end user of the underlying system studied. The guidelines must be grounded on the data analysis results presented in the report.

At the end of the conclusions section, students may wish to include a brief outlook on interesting future research. This would indicate potential avenues for future research that have not been pursued to any significant extent in the present study but may be of interest for future exploration.

- **Bibliography:** plagiarism is viewed as a serious offence, and you should take great care to cite all the sources used in your project.
- **Appendices:** The inclusion of appendices is not encouraged. However, if necessary, appendices should go after the bibliography. Appendices may be used for technical material or additional results that do not belong within the main text of the report. You do not need to include a copy of any software code that you have developed in your project. Instead, you may refer to a GitHub repository that you have used for your project. You may

include snippets of your software code in the main body of the report, if there is a good reason to do that.

Typesetting will typically be done in the “LaTeX” document preparation system, using the ‘report’ document class. Please see <http://www.latex-project.org/> for more information, including documentation. There are numerous online tutorials available for beginners (e.g. <http://www.latex-tutorial.com/tutorials/>). Microsoft Word is also acceptable, but make sure to use the Equation Editor for all equations.

Length and format of report

The report must be a maximum 50 A4 sides of paper including the bibliography and any appendices (but not including the title, contents and summary page). Anything beyond the 50th page will be ignored. The page limit is set to ensure that only relevant material is included. In particular, tables and diagrams should be included only when they are referred to somewhere in the text. Any devices used to try to get around the page limit will result in marks being deducted; please do not use non-standard fonts or shrunken diagrams.

You may want to avoid the temptation of submitting a Capstone project report that is much shorter than 40 pages. While a longer report is not always more convincing, you should allow yourself to argue your points fully. Shorter reports, unless they have a complete feel to them, often provoke questions of why particular extra material has not been included.

The report should be on A4 paper. The font size must be at least 11pt. The line spacing should be at least 1.5. The left-hand margin should be 3.5cm and the right-hand margin 2.0cm. Pages should be numbered; use Roman numerals (i, ii,...) for the title page, contents page and summary; use Arabic numerals (1, 2,...) for the remainder of the report starting with 1 on the first page of the introduction. Sections should be numbered consecutively, starting with 1. Introduction. Start new sections at the top of a new page. Tables and diagrams should be numbered. The required layout is easy to achieve in the Latex system.

Pencilled diagrams are not acceptable. Do not use footnotes. Take care of style, grammar, punctuation and general presentation. In summary, the format should be as follows:

- title, contents, summary + maximum 50 A4 sides
- font size minimum 11pt,
- line spacing minimum 1.5,
- left-hand margin 3.5cm, right-hand margin 2cm,
- title, contents, summary -- numbered using i, ii,...,
- the remainder of the report -- numbered using 1, 2,...,
- number sections, new sections start on new pages,
- number tables and diagrams,
- no footnotes.

References and bibliography

For references use the ‘Harvard’ or bracket system within the text. For example:

- "De Jong and Shephard (1995) put forward...",
- "Several authors (De Jong, 1989; Kohn and Ansley, 1989; Koopman, 93) make use of a..."
- "Harvey (1993, p.25) says, ‘Whether or not a mixed process is stationary depends solely on its autoregressive part’, but..."

In the bibliography, list works cited in alphabetical order by author. Give the author's surname, initials, year of publication, and title of work. For books, give the place of publication and publisher; for journal articles give the name of the journal, volume and pages; for articles in published collections, give the name of the editor, title of book, place of publication and publisher; for online sources give the source organisation, the URL and the date accessed. The appropriate use of italics and bold is important.

Examples of citations of (in order) a book, a journal article, and a conference proceedings paper are:

- Hastie, T. and Tibshirani, T. (2017). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (2nd edition). Springer Series in Statistics.
- Yang, Y., Wainwright, M. J. and Jordan, M. I. (2016). On the computation complexity of high-dimensional Bayesian variable selection. *Annals of Statistics*, Vol. 44, No. 6, pp 2497-2532.
- Dean, J. and Ghemawat (2004). MapReduce: simplified data processing on large clusters. *Proc. of OSDI'04*, San Francisco, CA, pp 10-10.

Again, this is straightforward to achieve in the Latex system (using BibTeX). EndNote is a useful software tool for managing your sources. References entered into EndNote can be exported into BibTex format (or imported directly into Word).

Note that for online sources referencing styles are not very well established, but it is essential to cite any material you have used. Be careful to use only reputable sources -- there is a lot of unreliable information on the web.

Plagiarism

Plagiarism is taking someone else's work or ideas and passing them off as your own. This arises in coursework as sections of text are lifted from books or Internet sources and submitted as the student's own work. It includes phrases or sentences taken from someone else work and mixed in with your own. Plagiarism is a very serious offence that is quite easy to detect. You are strongly encouraged to read widely and assimilate ideas from as many sources as possible. However, when you use other people's work you must give proper credit by providing a reference. Please note that this applies to any lecture materials you have used from LSE or elsewhere.

School regulations state "All work for classes and seminars, as well as scripts (which include, for example, examinations, essays, reports and any other work, including computer programs), must be the student's own work. The definition of a student's own work shall include work produced by collaboration expressly permitted by the department or institute concerned. Quotations must be placed properly within quotation marks and must be cited fully, and all paraphrased material must be acknowledged completely. Infringing this requirement, whether deliberately or not, or the deliberate or accidental passing off of the work of others as the work of the student is plagiarism."

Full details of disciplinary procedures in the case of plagiarism can be found at:

<https://info.lse.ac.uk/Staff/Divisions/Academic-Registrars-Division/Teaching-Quality-Assurance-and-Review-Office/Assets/Documents/Calendar/RegulationsAssessmentOffences-Plagiarism.pdf>

Individual Report Document

You are also required to write an individual report for your project of no more than 2 A4 sides in length (approximately 1,000 words). This should be submitted at the same time as your Capstone Project as a separate document. Further instructions will be provided nearer the time of submission.

Your individual report is a chance to apply critical thought to the process and lessons learned from the Capstone Project. For this reason, you might want to keep a research diary, keep notes on your experiences and difficulties, and have a detailed log of reflections that you can return to once the project is completed.

This document should have two components:

1. Individual technical contributions

This part should contain:

- A short introductory outline of the group's project, the main research problem or questions, its scope, and its most important findings and proposed recommendations;
- A summary of *your* individual technical contributions to the project (providing references to any measurable outcomes such as any particular software code that you have developed, and methodologies that you have individually identified and/or applied);
- An appreciation of what you think you could have done better;
- An identification of the areas for further personal development.

2. Working as a team member

This part should contain:

- A self-assessment of your individual contributions that helped other members of your project team to achieve more during the course of the project (providing references to any assistance that you gave to other team members and/or any lead roles that you had during the course of the project);
- An appreciation of *individual contributions of other team members*, summarising their individual contributions and how they helped you and/or other team members to achieve more;
- An identification of the areas for further personal development for other team members.

Your individual report document will be kept private to the extent that it will be used only by your supervisor and examiners to inform their decision-making in marking individual contributions. You are not expected to share this document with your Capstone colleagues, and it should not be included in the report that you give to your partner.

Your individual report should be submitted electronically via Moodle only. A hard copy is not required.

The Role of LSE Supervisors

You will have contact with your supervisor both at one-to-one meetings and the all-hands meetings.

Supervisors can be expected to offer advice on reading, guidance on the research approach and project structure, and some comments on the early draft sections.

Supervisors cannot be expected to correct spelling and grammar or to comment on a complete draft of the report.

You should have your first meeting with your supervisor towards the end of the Autumn term. You should then expect to meet approximately five times during the Winter term. It is your responsibility to contact your supervisor to arrange meetings. Please note that supervisors will not always be available outside term time, so you should make sure you know when your supervisor is going to be

available for face-to-face meetings and that you have their contact details for periods that they are not in the department. Formal supervision ends two weeks after the end of the Spring term. To get the most out of your supervisory meetings, it is a good idea to email your supervisor a brief progress update in advance of each meeting. It is also recommended that you write up a note after each meeting, summarising what was discussed and ideas to be followed up before the next meeting.

Organising your Time

Work on your project may be slowed down by the Spring exams. We advise that you, as a group, distribute the workload and organise your time to benefit the most from the Autumn and Winter terms. Below is a list of some of the things you should be working on these terms.

1. **Exploratory data analysis:** you should first focus on getting to know your dataset well and fully appreciating the research questions that you need to study. This will involve applying data manipulation and visualisation methods learned in courses. You may need to study a dataset given to you in a format that you have no prior experience working with (e.g. working with PostgreSQL and MongoDB databases). You are expected to be able to undertake any necessary steps to load and analyse your data.
2. **Research questions:** You should identify a list of four or five main research questions for your project. These may take the form of questions to answer, properties to establish or hypotheses to test. Your research questions will evolve as you work on your project; you should not feel bound by them. You are expected to pursue creative and critical thinking while defining and redefining your research questions. You should do a literature search to gain a broad understanding of related work.
3. **Methodology:** You should be proactive in identifying suitable methodology to address your problem. This may involve identifying the most appropriate machine learning methodology or statistical methodology to use to address your problem. This may involve doing a literature search to identify and learn about the most suited methodology. You should not be hesitant to learn about methodologies that you have not previously learned in school but are appreciated to be the ones you need for your project.
4. **Data analysis software:** You are expected to deploy a range of software tools that you learn throughout the programme as needed by your project. The exploratory data analysis would most likely involve developing Jupyter notebooks by using Python for analysis of data and using various libraries such as Numpy and Pandas for data analysis and Matplotlib for data visualisation. Your project may also require connecting to various databases such as the aforementioned PostgreSQL and MongoDB databases. Your project may also require working with large-scale datasets, in which case you may use software frameworks for distributed computing for big data that you learned in courses of the programme.

Each Capstone group is required to give a 10-minute presentation about their research at the first all-hands meetings. This should include an overview of the problem and key research questions, a description of the dataset and summary of exploratory data analysis, a summary of related work; and a summary of the research and a timetable for the remainder of the project.

By Easter, you should have:

- Performed a thorough exploratory data analysis;
- Gained a deep understanding of research questions that you study;
- Identified the methodology that you want to apply in your project;
- Conducted the literature review and collected information for the bibliography;
- Created a skeleton structure of chapters for your report;
- Made a plan of what you are going to do over the Spring.

Note that when you plan your work, it is important to start early (i.e. in the Autumn term) and be systematic. It is typically better to do a little each week than a lot in one week and then nothing for many subsequent weeks. You should have regular and frequent meetings with your supervisor. It is your responsibility to arrange meetings with your supervisor. You should also manage expectations when dealing with the Capstone corporate partner. This may involve maintaining progress and reporting about the progress at regular project meetings.

Capstone Marking

The final mark given to the student is based on two components: the final report mark and the individual contribution mark. These two components make the final mark with weights as follows:

1. **80% Capstone report** – two academic markers independently evaluate the report and then meet to agree on a final mark (one of these academic markers is the LSE supervisor).
2. **20% individual contribution** – at the end of the project, the supervisor aligns an individual contribution mark based on observed individual contributions to the project, overall engagement, and elements of teamwork, as well as based on the individual report document, which contains a self-assessment of individual contributions and a peer-assessment of individual contributions of other team members.

The peer assignment is conducted in an anonymous way where a Capstone Project team member provides an individual assessment for each other team member concerning a set of criteria such as the level of individual contribution and how much a team member has helped out team members to achieve more during the course of the project.

Capstone report marking scheme – Marks are assigned on a scale of 0-100%. Please see Page 12 for the marking rubric.

A selection of Capstone reports is viewed by an External Examiner, whose role is to ensure that the standards of internal marking are appropriate.

Capstone Project Marking Rubric

					Weighting
	Fail (0-49%)	Pass (50-59%)	Merit (60-69%)	Distinction (70% and over)	
1. Introduction and research question	Poor or not well-defined research question and poor summary of the report	Clear explanation of the research question and clear summary of the results	Clear explanation of the research question; well-motivated research question; clear summary of the results	Clear explanation of the research question; challenging and well-motivated research question; clear summary of the results; evidence of critical thinking	5%
2. Soundness of solution concepts (e.g. use of statistical methods, machine learning algorithms, computing)	Poor or inadequate application of theoretical concepts	Satisfactory application of theoretical concepts	Well identified and applied appropriate theoretical concepts	Well identified and applied theoretical concepts; shows deep understanding of the solution concepts; evidence of critical thinking	25%
3. Implementation	Poor implementation in software code	Basic implementation in software code applying existing functions	Good implementation in software code; appropriately applying existing functions and developing own if applicable for the given problem; neat and well commented code	Excellent implementation in software code demonstrating full understanding of how to code; developed own functions if applicable for the given problem; well-structured and commented code	25%
4. Choice and description of data	Poor choice of data and poor description of data properties	Satisfactory choice of data for the research question studied; good description	Good choice of data to address the research question, dataset has some complexity either with respect to structure or volume; clear description	Excellent choice of data to address the research question; data has complexity with respect to structure and volume; clear description	10%
5. Numerical evaluation	Uses inappropriate metrics to evaluate performance of the solution	Use of standard performance metrics	Good use of standard performance metrics allowing to evaluate and compare different solutions	Use of standard performance metrics following best practices to evaluate and compare different solutions; evidence of critical thinking	25%
6. Conclusion	No conclusion or not well justified concluding remarks	Overall sound conclusions	Sound conclusion, well-summarising main research results based on numerical evaluation	Excellent conclusion well-summarising main results of the study, strongly based on numerical evaluation results of the study; evidence of critical thinking	5%
7. Presentation quality	Poor structure with few explanations	Good structure of the report; cited references used in research	Clear presentation, well-structured report; cited references used in research	Clear presentation, well-structured report; effective communication of the results up to a professional standard; cited references used in research; cited references indicate a thorough research	5%

Individual Contribution Marking scheme

The individual mark has three elements:

1. **Individual technical contributions:** Has the individual made significant individual technical contributions? How do these individual technical contributions compare to those of other project team members? Has this individual contributed significantly more than other team members?
2. **Active participation and engagement in the project:** Did the individual actively participate and engage in the project, including attendance at project meetings. Did this individual have a lead role in various project activities such as presenting the project results and engaging in interactions with the project partner?
3. **Working as a team member:** Did the individual work well as a team member by taking their own share of the work responsibilities? Was the individual helping other team members to achieve more for the benefit of the project?

Capstone Evaluation Questionnaire

All students are required to complete and submit the Capstone Evaluation Questionnaire shortly after the submission of the Capstone Report (see Capstone Timeline).

The Capstone Evaluation results submitted by students will be used to inform the Capstone process in future year, namely, to identify good practice and to ensure we develop relevant Capstone projects pitched at the right level of technical difficulty; partner with responsive and engaging partner organisations; and provide adequate supervision and professional development support to students.

This questionnaire solicits student feedback on a number of issues:

- Student learning
- Level of difficulty of Capstone projects
- Group working
- Supervision
- Partner organisation

This is an independent piece of work and will be treated confidentially. The questionnaires will be processed by the MSc Programmes Manager and all the responses will be aggregated at a group level.

You are not expected to share this document with your Capstone colleagues, and it should not be included in the report that you give to your partner.

Capstone supervisors and academic markers will not have access to student responses until the marking process is complete.

Code of Conduct

All Capstone group members are expected to adhere to the following core principles of professional behaviour:

Be respectful

- Keep in touch with your Capstone colleagues
- Never allow disagreements over work to become personal
- Fully engage and actively participate in the project

Be professional

- Focus on the project: this may mean compromising your own views for the benefit of the project
- Identify and address any issues within the group before they have a chance to grow
- Keep in regular contact with your Capstone supervisor and other project team members

Be honest

- If other commitments (e.g. care of children/relatives, regular medical appointments) mean that you have restrictions on the times you can be available for the Capstone, discuss this openly with your colleagues from the outset

Be flexible

- Unforeseeable circumstances may arise which affect your project's development and/or members of your group: be prepared to make adjustments to enable your group to move forward; remember that this is a learning experience

Be realistic

- Keep a sense of perspective: the Capstone is one unit on the MSc Data Science programme
- When planning your Capstone work, take group members' commitments (including work for other courses) into account

Be kind

- Treat your Capstone colleagues as you would wish to be treated by others
- Be sensitive to different levels of knowledge and experience within your group